

QI ZHAO

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PROFESSIONAL SUMMARY

Medical systems engineer trained in the ISYMED (Medical Systems Engineering) track at ESIGELEC, with hands-on research experience applying deep learning to clinical imaging. Recent work at the Computer Vision Laboratory of the University of Würzburg focused on diffusion-model based SPECT image enhancement and multi-sensor synchronization for motion analysis. Comfortable across the full pipeline: embedded hardware, data processing, model development, and regulatory considerations for medical devices.

WORK HISTORY

05/2025 - 08/2025

Medical Image Enhancement By Using Diffusion Model

Computer Vision Laboratory of University of Würzburg - Würzburg

- Developed complete 3D medical-image enhancement framework integrating diffusion model DDIM with existing clinical SPECT reconstruction pipeline, in collaboration with Universitätsklinikum Würzburg, using provided papers, databases, and R2U-Net model. Modified existing R2U-Net with dual-channel structure, enhancing SPECT image detail optimization and blur recognition while preserving original architecture and evaluation flow. Integrated diffusion models with existing R2U-Net and PyTorch Lightning training infrastructure for improved adaptability.

03/2025 - 04/2025

Multi-event camera synchronization framework

Computer Vision Laboratory of University of Würzburg - Würzburg

- Designed and implemented multi-camera data acquisition framework enabling simultaneous control of heterogeneous imaging sensors, including DAVIS346 event cameras and DVSense devices. System supports synchronized triggering, data recording, real-time previews, and structured data storage for downstream motion and scene-change analysis.

SKILLS

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| <ul style="list-style-type: none">PythonCUDAOpenCVJupyter | <ul style="list-style-type: none">PyTorchArduinoMATLABLinux |
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EDUCATION

01/2022

ISYMED Dominant Engineering Diploma: MEDICAL SYSTEMS ENGINEERING

ESIGELEC - Rouen

- Robotique chirurgicale et téléopération (Surgical Robotics and Teleoperation): In this course, we researched and presented articles on medical technologies and AI in the form of PPT or oral presentations. I analyzed and discussed the key features of selected papers, explaining their significance. For the final assignment, I wrote a report on using CNN for Pneumonia Classification.
- ES Artificial Intelligence for Healthcare, Level 1 & 2: These courses introduced foundational AI concepts applied to healthcare, focusing on deep learning techniques, data analysis methods, and their practical applications in medical diagnostics and treatment.
- Notions générales en Machine Learning (Fundamentals of Machine Learning): Focused on core machine learning algorithms and their applications, providing an understanding of model training, feature selection, and data preprocessing.
- Management et plateaux techniques hospitaliers (Hospital Equipment Management and Technical Platforms): In this course, I visited local hospitals in Rouen to understand the sterilization and maintenance processes for surgical equipment, drug distribution management, and their role in healthcare operations.
- Dispositif médical: impact réglementaire dans le développement et la mise en œuvre d'un système embarqué (Medical Devices: Regulatory Impact on Embedded System Development and Implementation): Focused on the development and regulatory considerations of embedded systems in medical devices, including the modification, testing, and validation of an automated injection system.

LANGUAGES

Chinese NS

Native

French Delf B2

Advanced

English Toeic B2

Advanced